

Counting Inversions:

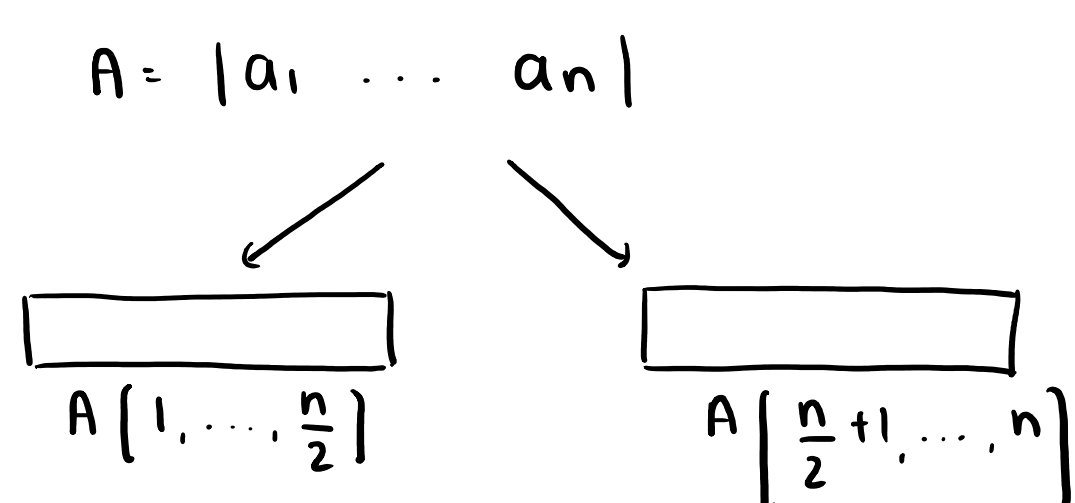
$$\left. \begin{array}{cccc} a_1 & a_2 & \dots & a_n \\ b_1 & b_2 & \dots & b_n \end{array} \right\} \cup a_i = \cup b_i$$

$\{a_1 \dots a_n\}$ not necessary that they are in sorted order.
 (i, j) st. $i < j$ and $a_i > a_j$ = INVERSION PAIR

of inversions $\cong$ measure of distance from sortedness.

$$\text{Det}(M) = \sum_{\sigma \in S_n} (-1)^{\text{sign}(\sigma)} \cdot \prod_{i=1}^n M_{i, \sigma(i)} \quad ? \quad [\text{Mahajan-Vinay, SODA 97}]$$

MERGE SORT:



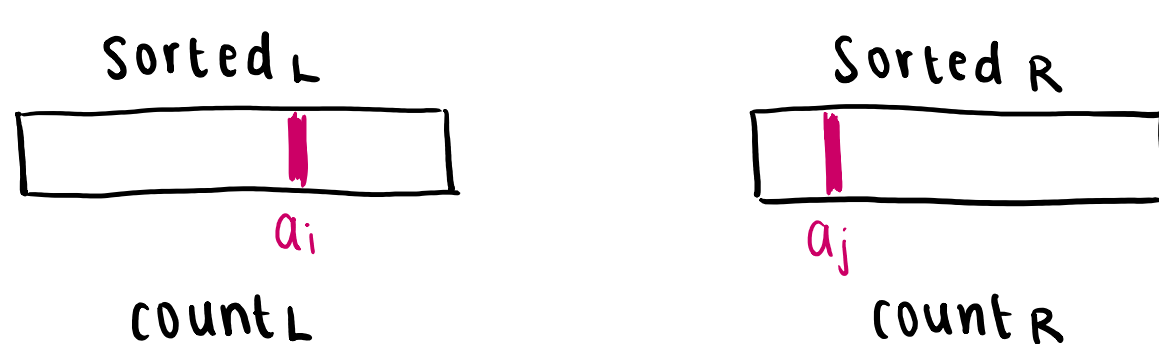
Q. Can we also count the no. of inversions while doing this?

Count Inversions ($A[1, \dots, n]$)

$P = \text{Count Inversions } A[1, \dots, \lfloor \frac{n}{2} \rfloor]$ } L

$Q = \text{Count Inversions } A[\lfloor \frac{n}{2} \rfloor + 1, \dots, n]$ } R

Brute force:
 $\binom{n}{2}$ pairs



$U = \#$ of inversions across L and R

Total no. of inversions = $P + Q + U$
 Brute force merge
 $O(n^2)$

Consider $a_i \in L$ and $a_j \in R$.

If $a_i > a_j$ then so are the elements to the right of a_i in Sorted L.



1-indexing
 $i = j = 1$ (init)

i and j are indicators of the heads of the lists.

$$\underline{S_L[i] > S_R[j]}$$

If yes, then $\text{count} += |S_L| - i + 1$
 $j \leftarrow j + 1$ } getting a count of inversions involving a_j

Else, $i \leftarrow i + 1$

$$T(n) = T\left(\left\lfloor \frac{n}{2} \right\rfloor\right) + T\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right) + O(n)$$

Median of Medians

= approximate median finding

Quicksort A:

$$T(n) = T(|L|) + T(|R|) + O(n)$$

↳ pivot p

Quicksort ($A_{<p}$) = L

Quicksort ($A_{>p}$) = R

Return L, {p}, R